above Earth’s surface, forms the “ozone layer,” a protective shield that absorbs most of the Sun’s ultraviolet (UV) radiation. Without it, more of the Sun's harmful radiation would reach the surface.

Since the mid-1970s, there has been a 4-percent decrease in the amount of ozone in the stratosphere. An even bigger decrease has been seen above the poles, particularly in spring. Over Antarctica, and the Arctic, the ozone layer is particularly depleted. This effect became known as “the ozone hole,” although it is better described as “the ozone depression,” since it is a thinning of the ozone layer rather than a complete hole. Nevertheless, the largest historical extent of the ozone “hole” occurred in September 2000, at 11 million sq miles (28.4 million sq km).

“Joe Farman [made] one of the most important geophysical discoveries of the 20th century.”

John Pyle and Neil Harris

Atmospheric scientists, University of Cambridge

Antarctic discovery

British geophysicist Joe Farman was one of the team who made the discovery in 1982. BAS teams had been collecting atmospheric data at the Halley Research Station in Antarctica since 1957. Their work was poorly funded, and they relied on dated instruments such as the Dobson meter—a rudimentary machine that worked properly only when wrapped in a duvet.

When Farman first noticed the drop in ozone levels, he found it hard to believe, and thought there must be a problem with his Dobson meter. He ordered a new meter for the next year—and it recorded an even bigger dip. The following year, the dip was bigger again. The year after, his team took their measurements 620 miles (1,000 km) from Halley. Again, there was a large dip. Farman decided it was time to publish, and in 1985 a paper by him, Brian Gardiner, and Jon Shanklin appeared in the journal *Nature*.